Rohit Kumar Singh

rohitkumarsingh82617@gmail.com | linkedin | github | LeetCode | +91 9798572504

Education

Noida Institute of Engineering and Technology — Noida, Uttar Pradesh
B.Tech in Computer Science Engineering (IOT) – SGPA: 8.52/10 — *Expected July 2027*

Sohan Lal Arya — Bokaro, Jharkhand
Higher Secondary School (PCM) – (86.6%) — *July 2022*

Projects

AlumHive – Alumni Connect Platform | *ReactJS, Java (Spring Boot), Bootstrap, JavaScript, REST APIs* | github

- Built a full-stack alumni networking platform to solve fragmented alumni-student communication using **ReactJS, Java Spring Boot, Bootstrap, and REST APIs**.
- Developed **role-based dashboards**, alumni directory, job board, mentorship system, and advanced search/filtering for **3 user roles**.
- Reduced alumni lookup time by **40%**, improved data retrieval through REST API integration, and delivered a fully responsive application across all devices.

Packet Analyzer (DPI Engine) | *C++, Multithreading, Networking, PCAP* | github

- Developed a Deep Packet Inspection engine capable of processing **1000+ packets/sec** using multithreading.
- Implemented packet parsing and SNI-based domain detection achieving **accurate application-level identification**.
- Built filtering system supporting **multiple rule types** (IP, domain, application-level blocking).
- Optimized performance to reduce latency by **30%** through efficient thread management.

Vector Database with RAG Pipeline | *C++, HNSW, KD-Tree, Ollama, REST API*

- Built a Vector Database from scratch in C++ implementing HNSW, KD-Tree, and Brute Force indexing for semantic search and retrieval.
- Optimized nearest-neighbor search using HNSW, achieving up to **10x faster query execution** than brute-force search while maintaining high retrieval quality.
- Developed a complete RAG pipeline using Ollama (`nomic-embed-text`, `llama3.2`), supporting **768-D embeddings**, REST APIs, document retrieval, and AI-powered question answering.

Technical Skills

Languages: C/C++, Python, Java, JavaScript, SQL

Frontend: ReactJS, NextJS, Bootstrap

Backend: FastAPI, REST APIs

Tools & Utilities: Git, Linux, Postman, VS Code

Databases: MySQL, SQLite

Achievements

- Solved 350+ problems on LeetCode, strengthening problem-solving and DSA skills

Certifications

- Java Programming Fundamentals | Infosys Springboard
- Introduction to Cybersecurity | Cisco Networking Academy
- Java Fundamentals | Oracle Academy
- AWS Academy Introduction to Cloud